

للأسف ما حصلنا أحد مصور أسئلة الاختبار بس واحد من الدكاترة
يعطيه العافية سوا مراجعة لاهم الأسئلة اللي ممكن تجي فالاختبار
وفادنا جدا
أتمنى يفيدكم.....

1. Download

بيجيك سؤال داو نلود نفس ذا:

Example (audio compression)

- How long would it take to download a 4 minute audio over a 56k bps. Assuming that the file is sampled at 44.1k Hz using 16 bit ADC?

of samples = $(4 \times 60) \times 44.1k = 10584k$ sample
Number of bits = $10584k \times 16 = 169344k$ bit
download duration = $169344k / 56k = 3024s = 50.4$ min

- How can we reduce the download duration?
Using audio compression

2. Error

بيجيك سؤال إرور نفس ذا:

Example

Detect and Correct errors in the following stream of bits knowing that the received stream has been encoded using rectangular coding

Received data stream

11000110 10100010 01101100

Solution

11000110 received correctly

10100010 received incorrectly

To correct the second block XOR the first block with the redundant block

11000110

01101100

10101010 therefore the error was in the fifth bit in the second block

3. Advantages and disadvantages of network:

Advantages:

- 1- Reduced hardware costs
- 2- Application sharing
- 3- Sharing information resources
- 4- Data management centralization
- 5- Connecting people

Disadvantages:

- 1- Loss of autonomy
 - 2- Lack of privacy
 - 3- Security threats
 - 4- Loss of productivity
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4. Differences between client-server and business-to-business (peer-to-peer) networks

Client-server: Made up of one or more file servers and clients, client software enables requests to be sent to the server, wired or wireless connections, do not slow down with heavy use.

business-to-business: Share files without a file server, easy to set up, best used for home or small offices with no more than 10 computers, do not require a network operating system, can be slow if there are too many users, security not strong

5. Name 3 types of LAN topologies, and how it works.

- 1- Bus topology: all the nodes including computers and servers are connected to a single cable termed as Bus, one node transmits at a time
- 2- Star topology: all the nodes are connected to Central Node. The devices are not connected to each other and transmits the messages to the Central Node
- 3- Ring topology: the nodes are interconnected to make a closed loop. Each node communicates with the nodes on its either side with the help of Token. The nodes with Token are allowed to transmit data

Multiple choice

Database:

6. **record** Includes a group of related fields.

7. One of the advantages of database management system **avoid redundancy**.

8. The **relation** appears in two-dimensional term.

9. **projection** creates relation in which each tuple has fewer attributes.

10. Determine data button helpful for making decision **data mining**.

Software Engineer:

11. The **waterfall** must be completed before next one began.

12. The **economical visibility** can be done with available financial resources.

13. What does it mean SDLC?

System development life cycle:

1- Plan

2- Analyze

3- Design

4- Implementation

يتم اكتشاف المشاكل في الخطوة الثانية (Analyze)

14. Single parity check

حيعطيك بلوكتين ويفلك ايت الصح ويعطيك 4 خيارات, اهم شي تفهم الطريقة.

15. One of the characteristic that distinguished the peer-to-peer network is **share file without a file server**.

16. One of the type of wired and wireless application is **internet telephony**.

True/False

17. CAN is a network of individual's own personal devices. **False**

18. The business-to- business network share the file with the server. **False**

19. The number of color is related to the number of bit assigned to each signal pixel. **True**

20. The bit depth of 8-bit, the total number of color in color plate is 128. **False**

21. Define digital signature and digital certificate.

digital signature: a mathematical scheme for verifying the authenticity of digital messages or documents.

digital certificate: an electronic document used to prove the validity of a public key

22. What is a parity bit?

a check bit, which is added to a block of data for error detection purposes

23. write one limitation (disadvantage) and one benefit (advantage) on single parity check.

Disadvantage: The receiver can detect that an error has occurred but it cannot determine exactly which bit is in error

Advantage: once error is detected the Receiver will inform the transmitter to re-transmit the block again

24. Two examples of graphical software of type bitmap.

1- Photoshop

2- GIMP

25. Explain how vector graphical picture work.

It creates images using a mathematical statement that places lines and shapes. Such as points, lines, curves, and polygons

26. Define honeypots.

computers baited (attracted) with fake data and purposely left vulnerable to study how intruders operate to prepare stronger defenses

27. Define key interception.

Occurs when a symmetric key encryption is stolen, allowing others to decrypt messages encrypted with that encryption key

28. SQL

بيجيبلك بلوكتين ويسالك هنا وش صار؟ انسيرت ولا ابديت؟

انت تشوف البلوكتين اذا لقيتهم نفس الشي **انسيرت**.

واذا الثانية مختلفة يعني **ابديت**.

لا تنسونا من دعائكم